

Yeast Fermentation and Fuel Sources

Teacher guide | Biology 2e, Chapter 7 | 45-60 minutes

Learning targets and timing

How does the available sugar affect gas accumulation in a yeast culture?

Compare baker's yeast with no added sugar, glucose, and sucrose. Record cumulative gas volume and compare average accumulation rates over the same time interval.

Suggested sequence: 8-10 min prediction and setup; 20-30 min observation; 12-20 min analysis. Virtual time can be accelerated. For photosynthesis and yeast, pool independent replicates from class groups; three sequential repeats need extra class time.

Route choices and sharing

In Teacher setup, select Virtual or Classroom for each treatment/field and create an assignment link. Assign all virtual for absences or a substitute. For a hybrid route, conduct one part in class and simulate the others; compare data-source limitations before pooling. These links are editable and do not restrict access.

Materials

- Baker's yeast, glucose, sucrose, water, balance, thermometer and timer
- Three matched flasks and tubing connected to freely moving 100 mL gas syringes
- Teacher-prepared 30 deg C water bath; matched mixtures at 50 mL final volume

Classroom procedure

1. Label matched setups no added sugar, glucose, and sucrose. Prepare each sugar mixture with 2 g sugar; the water control receives no added sugar.
2. Add 0.5 g yeast to each and adjust to 50 mL final volume. Use the same yeast batch and equilibration time. Maintain 30 deg C with a water bath.
3. Connect each flask to a freely moving gas syringe. Check for leaks and unobstructed motion. Record the initial gas reading and subtract it from subsequent readings.
4. Record cumulative gas at 0, 5, 10, ... 30 minutes. Do not compare foam height or balloon circumference with the app's milliliter values.
5. Repeat with fresh mixtures for trials 2 and 3. Pool independent trials from class groups for a single-period lab; sequential repeats require additional time. Compare volume change divided by elapsed time, retain outliers, and explain suspected problems.

Safety: Use baker's yeast only. Do not taste cultures. Never seal gas production in a rigid closed container or block the syringe plunger. Stop before apparatus capacity is reached. Teacher checks glassware and warm-water handling.

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Expected reasoning and troubleshooting

Virtual expectation: both sugar treatments accumulate more gas than the no-added-sugar control. Glucose and sucrose may exchange rank across trials. Encourage evidence-based comparison rather than a fixed "winning sugar."

Fermentation regenerates NAD⁺ so glycolysis can continue. Fermentation and anaerobic respiration are different pathways; yeast gas volume does not uniquely establish which metabolic processes occurred.

No-added-sugar is not necessarily zero because yeast has internal reserves and apparatus has background effects. A no-yeast control and leak/temperature checks strengthen the inference.

Model boundaries

Volumes are illustrative, generated curves with a lag and trial variation. Glucose and sucrose responses overlap; their ranking is not universal. The water control can show small background gas. The model does not establish anaerobic conditions or identify collected gas chemically.

Average gas accumulation rate = (final volume - initial volume) / elapsed minutes. Gas from yeast is consistent with CO₂ production, but volume alone does not prove fermentation rather than aerobic metabolism or directly measure ATP yield.

Suggested assessment - 12 points

Dimension	Evidence to look for	Points
Design	Reasoned prediction; variables, controls or sampling plan	0-3
Data	Complete observations, correct units, source labels, repeats	0-3
Analysis	Appropriate calculations and comparisons supported by data	0-3
Explanation	Biological reasoning, uncertainty, and a feasible improvement	0-3

For each dimension: 0 = absent; 1 = partial or major errors; 2 = mostly sound with a minor omission; 3 = complete and well supported. Suggested teacher rubric only; the app does not assign a grade.

Downloads, access, and classroom pilot

Student worksheet: three printable pages. Teacher guide: procedures, interpretation, and rubric. Reports and CSV data download from Analyze & explain. Teacher guides are public resources, not protected answer keys. Before classroom use, pilot the equipment, timing, and reading level. Editable Word and fillable PDF versions are planned.

Reading and method connections

OpenStax Biology 2e, 7.5: Open source

Text and diagrams in these labs are original teaching materials. Simulated observations are not measurements from the linked sources.